# National Science Olympiad 2011 Dynamic Planet Exam, Division B
# ANSWER KEY

**Questions 1-9:** 2 pts each

1. A
2. B
3. B
4. C
5. D
6. B
7. A
8. C
9. C

**Question 10:** 6 pts **TIE BREAKER 3**

10. 2 pts relief (1 pt highest, 1 point lowest elev)

(700 – (somewhere between 585-590) = somewhere between 115-110 ft)

2 pts distance in miles (1 pt correct measurement, 1 pt conversion)

(2 inches * (62,500 in/ 1 in )*(1 ft/ 12in)*(1mi/5280ft) = 125,000in = 10,416.7 ft = 1.973 miles )

2 pts final calc/ answer (1 pt correct equation, 1 pt correct final answer)

(somewhere between 110-115 ft / 1.973 miles = somewhere between 55.76 – 58.29 ft/mile)

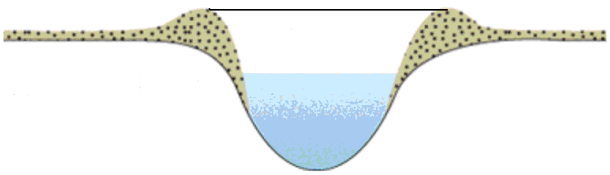
**Question 11:** 3 pts **TIE BREAKER 4**

11. ½ pt each blank (if incorrect number filled in blank only ¼ pt given)

| | |
|---|---|
| ___ increased evapotranspiration | _1_ establishment of meanders |
| _2_ lowered local base level | ___ terrace formation |
| _3_ increased vertical erosion | ___ high dissolved load |

**Questions 12-17:** 2 pts each

12. C
13. B
14. D
15. D
16. B
17.



**Question 18:** 4 points

18. 1 pt conversion (45 cm/s * 1m/100cm = 0.45 m/s)

    1 pt area (7m * 2m = 14 $m^2$)

    2 pts Q calc/answer (1 pt correct equation, 1 pt correct answer)

    (0.45 m/s * 14 $m^2$ = 6.3 $m^3$/s)

**Question 19:** 8 points **TIE BREAKER 1**

1 pt labeled contours

1 pt correct number of contours and interval

4 pts that elevation points are in-between correct contour lines (¼ pt each point)

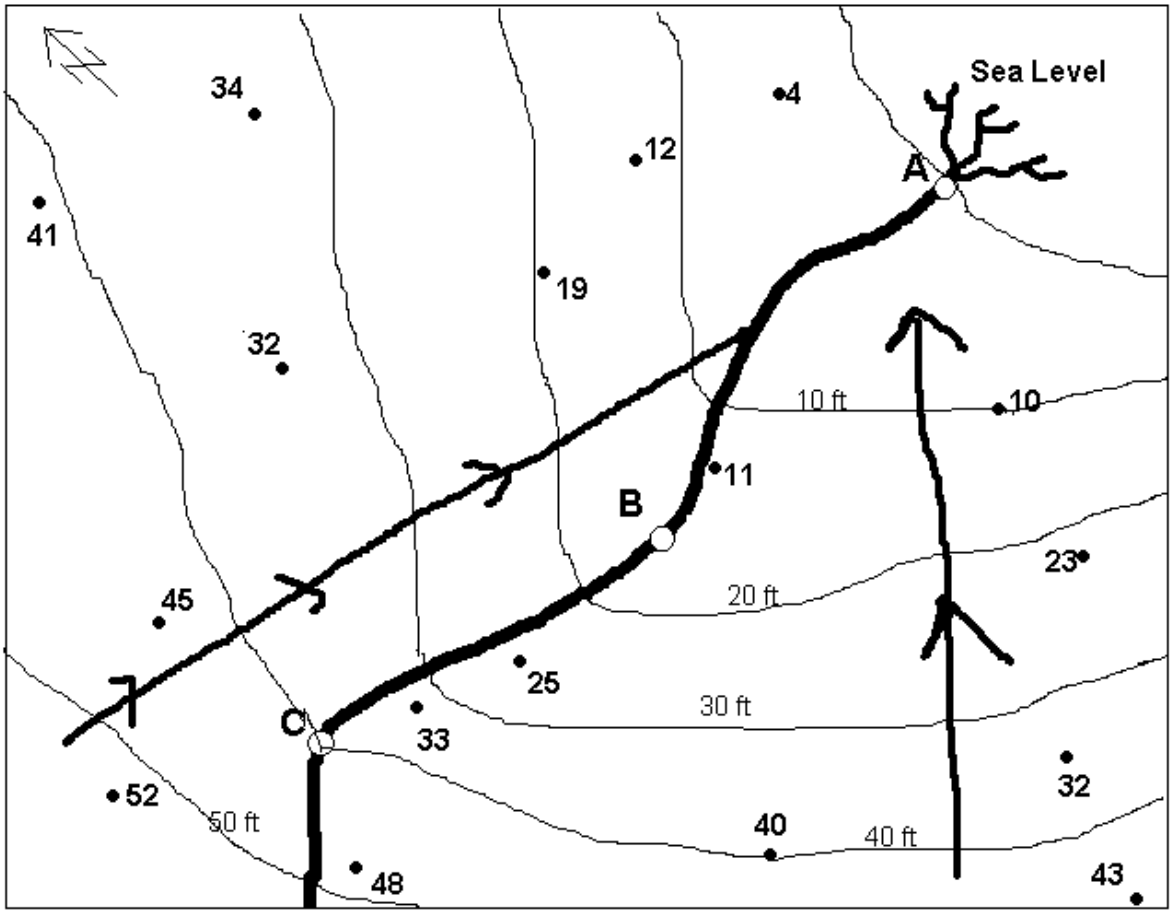
2 pts general shape correct

**Question 20:** 4 points

2 pts each line:

1 pt correct arrow direction

1 pt perpendicular to contours



**Question 21:** 3 points (1 pt each)

A: between 0-2 ft

B: between 13-16 ft

C: between 39-41 ft

**Questions 22-23:** 2 pts each

22. B

23. C

**Question 24:** 6 points (3 pts in, 3 pts out) -½ pt for each incorrect answer

Inputs: Precipitation, Groundwater, Runoff/Overland flow

Outputs: Evaporation, Groundwater

**Question 25:** 4 points (1 pt each blank)

A: summer

B: winter

C: fall

D: spring

**Questions 26-27:** 2 pts each

26. agricultural runoff (pesticides and/or fertilizers), acid rain, septic tank discharge

27. A

**Questions 28-30:** 4 pts each (1 pt each box) **TIE BREAKER 2**

-½ pt if don't list all communities that are affected

28.

| **Positive Impacts of the Dam** | **Communities this will affect** |
|---|---|
| *reservoir creation for recreation (boating, fishing, etc) or water supply | *potentially all of them, mostly A as it is the closest |
| *enhance navigation | *all or A |
| *could provide electricity | *all or A |
| *flood control | *A |

| **Negative Impacts of the Dam** | **Communities this will affect** |
|---|---|
| *severe erosion downstream due to loss of sediment in reservoir | * mostly B (maybe farmers, C) |
| *reduction in sediment supply to delta/beaches | *C |
| *affects fish/plant habitats, loss of diversity | *all |

29.

| **Positive Impacts of the Levees/Support Structures** | **Communities this will affect** |
|---|---|
| *levees- protect community B from flooding | *B |
| *support structures- keep meander, and thus B from being cutoff, also scar can't form reducing property values/loss of water front, etc. | *B |

| **Negative Impacts of the Levees/Support Structures** | **Communities this will affect** |
|---|---|
| *increase flooding risk downstream | *farmers, C |
| *navigation more difficult around meander | *all, esp. A for getting to ocean |

30.

| **Positive Impacts of draining wetlands** | **Communities this will affect** |
|---|---|
| additional agricultural land for more food supply | all, everyone eats food! |

| **Negative Impacts of draining wetlands** | **Communities this will affect** |
|---|---|
| *affects habitats, loss of biodiversity | *all |
| *increased flooding | *C, farmers |
| *decreased water quality | *C |
| *decrease carbon storage | *all |

# Science Olympiad National Tournament - Repaired practice edition - Answer key

### Division B | Dynamic Planet | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

### ANSWER KEY - Review only after submitting.

**repair-case. (2 points) Authored written substitute for unavailable source material: Terrace elevation alone establishes exact age?**

Unit: Fluvial landforms and river history.
Worked-case data: Two events share peak discharge but one lasts longer. Why might their geomorphic effects differ?

A. Only for sandstone

B. Only below sea level

C. No

D. Always

**Correct: C**